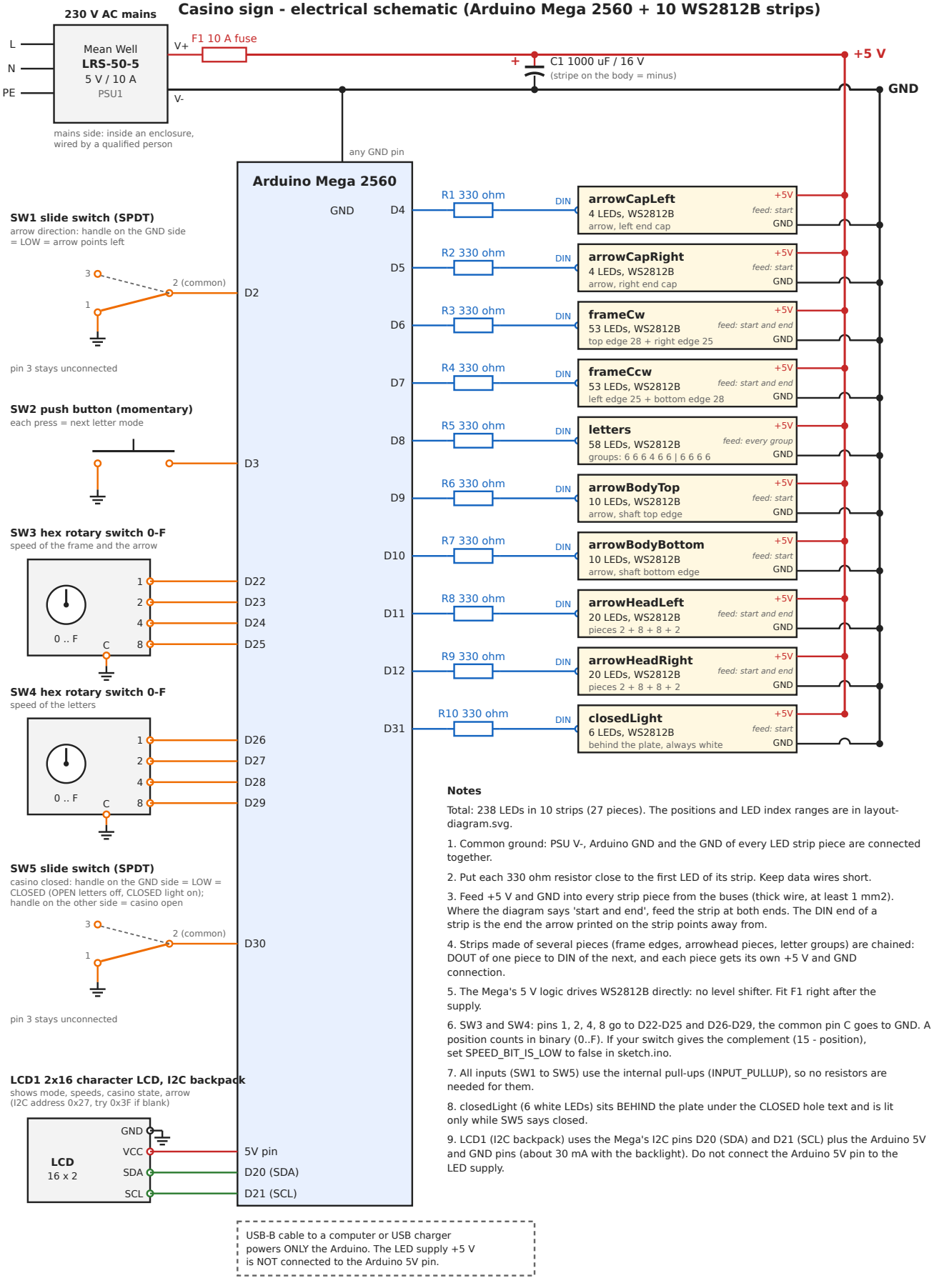
Casino sign - electrical schematic (Arduino Mega 2560 + 10 WS2812B strips)



Notes

Total: 238 LEDs in 10 strips (27 pieces). The positions and LED index ranges are in layout-diagram.svg.

1. Common ground: PSU V-, Arduino GND and the GND of every LED strip piece are connected together.

2. Put each 330 ohm resistor close to the first LED of its strip. Keep data wires short.

3. Feed +5 V and GND into every strip piece from the buses (thick wire, at least 1 mm2). Where the diagram says 'start and end', feed the strip at both ends. The DIN end of a strip is the end the arrow printed on the strip points away from.

4. Strips made of several pieces (frame edges, arrowhead pieces, letter groups) are chained: DOUT of one piece to DIN of the next, and each piece gets its own +5 V and GND connection.

5. The Mega's 5 V logic drives WS2812B directly: no level shifter. Fit F1 right after the supply.

6. SW3 and SW4: pins 1, 2, 4, 8 go to D22-D25 and D26-D29, the common pin C goes to GND. A position counts in binary (0..F). If your switch gives the complement (15 - position), set SPEED_BIT_IS_LOW to false in sketch.ino.

7. All inputs (SW1 to SW5) use the internal pull-ups (INPUT_PULLUP), so no resistors are needed for them.

8. closedLight (6 white LEDs) sits BEHIND the plate under the CLOSED hole text and is lit only while SW5 says closed.

9. LCD1 (I2C backpack) uses the Mega's I2C pins D20 (SDA) and D21 (SCL) plus the Arduino 5V and GND pins (about 30 mA with the backlight). Do not connect the Arduino 5V pin to the LED supply.